

Table 5: **Results for 2.8B parameter models trained on repeated data on OSCAR for 55B total tokens.** Scores are normalized averages of 0-5 few-shots and reported as percentages. We report mean/std. err. across five different models, each trained with a different random seed.

Data Budget	55B	28B	18B	14B	11B	9B	4B	1.25B
Epochs	1	2	3	4	5	7	14	44
ANLI R1	-0.3 ± 0.5	-0.6 ± 1.3	0.2 ± 0.6	0.3 ± 1.1	0.2 ± 1.2	-0.1 ± 1.1	-0.1 ± 0.5	-0.7 ± 1.3
ANLI R2	1.0 ± 1.0	1.1 ± 0.3	1.7 ± 0.8	2.3 ± 0.8	1.4 ± 1.0	0.8 ± 0.7	1.0 ± 0.7	2.3 ± 0.7
ANLI R3	0.4 ± 0.8	0.5 ± 0.5	-0.2 ± 0.8	-0.1 ± 1.0	1.1 ± 0.6	0.7 ± 0.4	-0.2 ± 0.9	0.5 ± 1.2
ARC-Challenge	-1.4 ± 0.8	-0.6 ± 0.8	-1.7 ± 0.1	-1.6 ± 0.7	-1.6 ± 0.6	-1.4 ± 0.5	-1.9 ± 0.8	-5.0 ± 1.1
ARC-Easy	39.7 ± 0.3	39.6 ± 0.8	39.5 ± 0.6	39.3 ± 0.5	38.7 ± 0.6	38.7 ± 0.4	36.9 ± 0.4	25.4 ± 0.7
BoolQ	12.8 ± 4.4	7.8 ± 3.8	7.9 ± 3.8	3.3 ± 5.4	0.2 ± 3.0	2.3 ± 6.1	-2.1 ± 2.4	7.4 ± 6.1
CB	19.7 ± 5.1	15.4 ± 7.3	13.2 ± 5.1	12.6 ± 2.6	21.7 ± 3.6	15.4 ± 3.7	16.2 ± 5.2	9.7 ± 5.7
COPA	42.7 ± 2.2	39.5 ± 2.2	40.9 ± 2.0	41.5 ± 2.1	38.5 ± 2.4	40.4 ± 2.4	38.6 ± 2.6	28.5 ± 3.1
HellaSwag	16.3 ± 0.1	16.3 ± 0.2	16.3 ± 0.2	16.1 ± 0.2	16.0 ± 0.1	15.9 ± 0.2	15.0 ± 0.2	11.7 ± 0.1
PiQA	41.2 ± 0.7	41.4 ± 0.5	40.3 ± 0.4	40.6 ± 0.5	40.3 ± 0.9	39.8 ± 0.6	38.8 ± 1.1	31.0 ± 0.4
RTE	3.9 ± 1.1	2.1 ± 1.6	2.3 ± 3.3	1.6 ± 3.0	0.5 ± 2.1	2.9 ± 2.5	0.9 ± 3.4	-3.2 ± 2.7
SciQ	83.2 ± 0.6	82.4 ± 0.6	82.1 ± 0.9	82.6 ± 0.7	81.5 ± 0.9	80.5 ± 0.6	76.5 ± 1.3	57.7 ± 1.8
StoryCloze 2016	52.8 ± 0.3	52.9 ± 0.4	52.6 ± 0.3	53.0 ± 0.4	52.3 ± 0.4	52.4 ± 0.4	51.8 ± 0.7	47.9 ± 0.5
WinoGrande XL	5.8 ± 0.9	4.4 ± 1.4	4.5 ± 0.3	4.2 ± 1.3	4.5 ± 0.6	4.1 ± 0.7	1.7 ± 1.2	0.8 ± 1.3
E2E NLG	20.3 ± 0.3	19.9 ± 0.5	19.9 ± 0.7	20.9 ± 0.9	19.7 ± 0.7	20.4 ± 0.6	19.1 ± 0.8	14.2 ± 0.7
XSUM	3.0 ± 0.1	2.9 ± 0.0	2.9 ± 0.3	2.9 ± 0.2	2.9 ± 0.1	2.8 ± 0.3	2.6 ± 0.2	1.8 ± 0.1
WebNLG EN	8.8 ± 0.4	8.3 ± 0.6	8.5 ± 0.3	8.4 ± 0.6	8.1 ± 0.2	8.2 ± 0.2	7.2 ± 0.3	3.3 ± 0.3
WikiLingua EN	2.9 ± 0.1	3.1 ± 0.2	3.1 ± 0.1	3.0 ± 0.1	3.1 ± 0.1	3.2 ± 0.3	2.7 ± 0.2	1.7 ± 0.2
bAbI	15.5 ± 1.0	15.7 ± 1.1	15.3 ± 0.8	15.1 ± 1.5	15.9 ± 1.1	16.2 ± 0.9	14.3 ± 0.6	6.6 ± 0.6
Average	19.4 ± 0.5	18.5 ± 0.2	18.4 ± 0.4	18.2 ± 0.4	18.2 ± 0.4	18.1 ± 0.4	16.8 ± 0.5	12.7 ± 0.7

Table 6: **Results for 4.2B parameter models trained on repeated data on C4 for 84B total tokens.** Scores are normalized averages of 0-5 few-shots and reported as percentages. We report mean/std. err. across five different models, each trained with a different random seed.

Unique Tokens	84B	42B	28B	21B	17B	12B	6B	1.9B
Epochs	1	2	3	4	5	7	14	44
ANLI R1	-1.0 ± 0.3	-0.7 ± 1.1	-0.7 ± 1.0	-0.4 ± 1.1	0.4 ± 0.8	0.5 ± 1.1	0.1 ± 0.9	0.2 ± 0.9
ANLI R2	0.8 ± 0.5	0.8 ± 0.8	0.0 ± 1.4	0.5 ± 0.7	0.5 ± 0.9	0.3 ± 1.0	0.7 ± 0.7	2.5 ± 1.0
ANLI R3	1.1 ± 0.7	0.8 ± 0.9	0.3 ± 0.8	1.4 ± 1.1	1.3 ± 0.9	2.3 ± 0.2	1.3 ± 0.2	1.6 ± 1.2
ARC-Challenge	5.3 ± 0.6	5.1 ± 0.9	5.2 ± 2.0	6.0 ± 0.8	4.7 ± 0.8	3.1 ± 0.4	2.9 ± 1.0	-1.3 ± 1.0
ARC-Easy	49.2 ± 0.9	50.4 ± 1.2	47.4 ± 4.5	49.4 ± 0.7	48.7 ± 1.5	44.9 ± 0.7	45.0 ± 1.2	31.9 ± 0.9
BoolQ	18.2 ± 4.0	19.6 ± 5.1	22.1 ± 1.0	20.4 ± 3.6	18.4 ± 6.0	18.4 ± 3.9	18.9 ± 2.6	-3.3 ± 7.1
CB	12.0 ± 7.2	8.5 ± 9.2	7.9 ± 10.4	19.6 ± 7.3	17.8 ± 7.3	15.1 ± 5.8	17.5 ± 3.5	19.5 ± 6.6
COPA	59.1 ± 5.4	57.7 ± 3.5	56.7 ± 2.0	55.5 ± 2.4	56.8 ± 1.8	58.9 ± 1.7	48.7 ± 3.3	34.9 ± 3.4
HellaSwag	27.8 ± 4.8	30.2 ± 0.5	29.8 ± 0.9	29.9 ± 0.7	28.5 ± 1.1	29.0 ± 0.5	27.0 ± 1.2	19.7 ± 0.5
PiQA	50.6 ± 0.5	50.8 ± 0.5	48.6 ± 3.4	50.9 ± 0.7	50.3 ± 1.3	49.5 ± 0.4	47.6 ± 1.2	39.5 ± 1.3
RTE	5.6 ± 3.1	2.6 ± 3.9	7.2 ± 2.7	7.0 ± 3.2	8.8 ± 5.3	9.3 ± 3.6	3.0 ± 4.3	2.6 ± 4.2
SciQ	84.6 ± 3.9	86.1 ± 1.3	84.4 ± 3.7	85.9 ± 0.7	86.2 ± 0.8	79.0 ± 0.7	81.1 ± 1.4	65.3 ± 1.1
StoryCloze 2016	61.1 ± 3.7	62.6 ± 0.2	61.9 ± 2.2	62.6 ± 0.4	61.8 ± 0.8	61.5 ± 0.8	60.1 ± 0.7	53.9 ± 0.5
WinoGrande XL	17.0 ± 2.6	17.8 ± 1.4	16.5 ± 1.8	17.1 ± 1.8	14.9 ± 1.5	15.9 ± 1.2	11.8 ± 1.5	3.9 ± 0.8
E2E NLG	18.2 ± 1.2	18.8 ± 0.8	17.8 ± 1.5	16.0 ± 2.2	15.9 ± 2.5	13.8 ± 1.3	15.7 ± 0.9	11.2 ± 1.4
XSUM	2.9 ± 0.2	3.0 ± 0.2	2.8 ± 0.3	2.9 ± 0.2	2.9 ± 0.2	1.0 ± 0.4	2.4 ± 0.1	1.8 ± 0.1
WebNLG EN	4.8 ± 2.0	5.7 ± 0.2	5.4 ± 0.3	5.6 ± 0.2	5.4 ± 0.5	5.5 ± 0.1	5.4 ± 0.2	3.4 ± 0.3
WikiLingua EN	3.3 ± 0.5	3.6 ± 0.1	3.4 ± 0.1	3.4 ± 0.1	3.3 ± 0.1	1.4 ± 0.6	3.0 ± 0.1	2.2 ± 0.1
bAbI	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0
Average	22.1 ± 1.7	22.3 ± 0.9	21.9 ± 1.2	22.8 ± 0.5	22.5 ± 0.6	21.6 ± 0.5	20.6 ± 0.6	15.2 ± 1.0

Table 7: **Results for 4.2B parameter models trained on repeated data on OSCAR for 84B total tokens.** Scores are normalized averages of 0-5 few-shots and reported as percentages. We report mean/std. err. across five different models, each trained with a different random seed.

Unique Tokens	84B	42B	28B	21B	17B	12B	6B	1.9B
Epochs	1	2	3	4	5	7	14	44
ANLI R1	-0.9 ± 0.5	-0.8 ± 1.1	-0.9 ± 1.4	-0.4 ± 0.4	-0.1 ± 1.2	0.3 ± 1.1	-0.5 ± 0.8	1.1 ± 1.3
ANLI R2	0.7 ± 0.9	0.7 ± 1.1	1.3 ± 1.0	1.5 ± 1.1	1.7 ± 1.3	0.9 ± 0.8	0.9 ± 1.0	1.7 ± 1.3
ANLI R3	0.4 ± 0.6	0.6 ± 0.4	0.7 ± 0.3	0.4 ± 0.8	0.7 ± 1.2	0.6 ± 1.2	0.7 ± 0.5	0.8 ± 1.2
ARC-Challenge	1.3 ± 0.5	1.8 ± 0.5	1.6 ± 0.7	2.4 ± 1.1	1.6 ± 0.7	2.0 ± 0.7	1.6 ± 0.5	-2.1 ± 0.5
ARC-Easy	45.5 ± 0.8	45.1 ± 1.2	44.8 ± 0.9	44.8 ± 0.6	45.0 ± 1.0	43.9 ± 0.7	40.7 ± 0.7	28.0 ± 0.9
BoolQ	14.5 ± 1.9	15.1 ± 4.6	10.8 ± 5.1	12.5 ± 1.9	6.7 ± 4.0	10.1 ± 4.2	-0.0 ± 6.9	-4.3 ± 7.2
CB	21.3 ± 2.3	19.2 ± 3.8	12.9 ± 6.4	16.9 ± 3.4	15.1 ± 9.4	17.8 ± 3.6	15.0 ± 8.1	11.2 ± 4.1
COPA	43.1 ± 3.0	42.5 ± 3.7	44.4 ± 1.1	43.0 ± 3.4	41.8 ± 2.3	44.6 ± 2.7	40.3 ± 3.0	34.9 ± 4.9
HellaSwag	21.1 ± 0.2	21.0 ± 0.2	20.9 ± 0.1	20.7 ± 0.2	20.5 ± 0.3	20.3 ± 0.1	19.3 ± 0.1	14.5 ± 0.2
PiQA	45.3 ± 0.9	44.8 ± 0.7	44.8 ± 0.9	44.4 ± 0.6	44.3 ± 0.6	43.9 ± 0.5	42.2 ± 0.9	34.0 ± 0.8
RTE	4.2 ± 2.8	1.5 ± 2.4	-1.1 ± 3.9	-2.5 ± 3.9	5.3 ± 1.8	4.4 ± 1.9	1.6 ± 2.2	-1.0 ± 2.4
SciQ	86.6 ± 0.7	86.5 ± 0.5	86.0 ± 0.2	86.3 ± 1.0	85.4 ± 0.8	84.7 ± 0.4	82.0 ± 1.4	62.9 ± 2.5
StoryCloze 2016	56.5 ± 0.6	56.8 ± 0.6	56.5 ± 0.7	55.8 ± 0.3	55.9 ± 0.2	56.0 ± 0.3	54.5 ± 0.7	49.3 ± 0.2
WinoGrande XL	9.7 ± 1.4	9.0 ± 1.8	9.5 ± 0.7	8.9 ± 1.0	7.8 ± 1.2	7.4 ± 1.4	6.8 ± 1.4	2.1 ± 1.0
E2E NLG	21.4 ± 1.3	21.9 ± 0.4	21.2 ± 1.0	21.8 ± 0.6	21.0 ± 0.9	20.5 ± 0.7	20.9 ± 1.0	16.0 ± 0.6
XSUM	3.6 ± 0.2	3.5 ± 0.2	3.5 ± 0.2	3.5 ± 0.2	3.5 ± 0.3	3.2 ± 0.5	3.0 ± 0.2	1.9 ± 0.1
WebNLG EN	9.9 ± 0.4	9.7 ± 0.8	9.3 ± 0.6	9.7 ± 0.5	9.3 ± 0.7	9.4 ± 0.3	8.9 ± 0.5	3.8 ± 0.4
WikiLingua EN	3.9 ± 0.1	3.8 ± 0.2	3.6 ± 0.3	3.7 ± 0.2	3.6 ± 0.2	3.7 ± 0.1	3.3 ± 0.2	2.1 ± 0.2
bAbI	15.0 ± 7.5	19.0 ± 1.2	18.8 ± 1.4	18.5 ± 1.4	19.2 ± 0.6	18.1 ± 1.4	14.5 ± 1.5	9.6 ± 1.7
Average	21.2 ± 0.2	21.1 ± 0.4	20.4 ± 0.3	20.6 ± 0.5	20.4 ± 0.5	20.6 ± 0.2	18.7 ± 0.5	14.0 ± 0.6

Table 8: **Results for 8.7B parameter models trained on repeated data on C4 for 178B total tokens and a data-constrained compute-optimal 6.3B model.** Scores are normalized averages of 0-5 few-shots and reported as percentages. The two models with 25 billion unique tokens are the ones depicted in Figure 1 (right). The data-constrained compute-optimal variant (6.3 billion parameters) performs better by using fewer parameters and repeating more data.

Parameters	8.7B								6.3B
Unique Tokens	178B	88B	58B	44B	35B	25B	13B	4B	25B
Epochs	1	2	3	4	5	7	14	44	9.7
ANLI R1	-0.9	-1.2	-4.2	0.7	-1.3	0.1	1.2	2.1	-0.9
ANLI R2	-0.4	-1.2	-0.2	0.2	-0.4	-0.1	0.4	2.2	1.0
ANLI R3	0.7	0.5	0.7	1.8	0.4	1.6	2.0	4.0	2.6
ARC-Challenge	12.2	11.9	10.5	12.2	10.6	11.8	8.3	2.2	12.7
ARC-Easy	58.5	58.0	56.9	57.4	56.7	58.5	52.9	37.4	57.2
BoolQ	26.1	31.8	31.3	30.3	28.8	28.5	27.9	4.1	30.6
CB	7.6	12.9	-15.2	17.9	14.3	-22.8	-12.1	17.4	6.2
COPA	68.0	64.7	62.3	66.3	63.3	70.0	57.0	45.0	66.0
HellaSwag	37.8	37.8	37.3	37.4	37.1	37.5	36.1	27.5	38.1
PiQA	55.9	55.6	54.7	56.5	55.8	53.9	52.4	45.7	54.3
RTE	14.1	11.4	11.0	8.7	15.9	-2.6	-1.8	-3.2	7.7
SciQ	90.4	91.1	90.7	90.0	89.8	89.8	87.9	72.9	90.3
StoryCloze 2016	68.3	67.3	67.2	67.6	67.8	66.8	66.2	58.9	68.4
WinoGrande XL	26.3	27.7	26.5	29.0	26.1	23.5	18.1	10.0	27.0
E2E NLG	20.5	17.9	18.7	20.0	17.2	17.7	17.4	11.2	16.9
XSUM	3.6	3.3	3.8	3.8	3.5	3.0	3.3	2.0	3.8
WebNLG EN	5.3	5.8	5.9	5.6	5.8	5.2	5.7	4.9	5.3
WikiLingua EN	4.1	4.2	4.2	4.1	4.2	4.0	3.5	2.7	4.0
bAbI	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Average	26.2	26.3	24.3	26.8	26.1	23.5	22.4	18.3	25.9

Table 9: **Results for 8.7B parameter models trained on repeated data on OSCAR for 178B total tokens.** Scores are normalized averages of 0-5 few-shots and reported as percentages.

Unique Tokens	178B	88B	58B	44B	35B	25B	13B	4B
Epochs	1	2	3	4	5	7	14	44
ANLI R1	-1.3	-2.3	-0.5	-1.8	0.1	-0.3	2.6	-0.4
ANLI R2	0.8	3.2	-0.2	-1.3	1.0	0.2	1.5	0.5
ANLI R3	1.1	1.2	1.3	0.9	2.8	-0.4	1.1	-0.1
ARC-Challenge	6.9	6.7	6.9	3.8	6.6	4.8	4.0	-0.9
ARC-Easy	50.2	51.6	51.2	51.0	51.9	50.8	47.0	33.0
BoolQ	18.4	11.7	19.4	22.4	17.5	20.8	7.6	4.1
CB	11.2	13.4	16.1	19.6	21.4	25.0	9.8	20.1
COPA	46.7	53.0	52.0	53.7	51.0	53.3	48.7	41.7
HellaSwag	27.4	27.2	26.8	26.8	27.3	26.7	25.5	19.6
PiQA	49.2	49.3	50.1	48.7	48.1	47.2	45.6	37.0
RTE	-0.5	1.1	0.2	1.2	10.2	3.2	-3.0	-7.8
SciQ	88.1	88.0	88.4	87.9	87.9	87.4	86.3	64.6
StoryCloze 2016	61.6	61.1	60.2	60.6	61.3	59.0	58.8	52.7
WinoGrande XL	17.6	16.3	15.4	13.7	13.9	12.8	10.8	-0.6
E2E NLG	23.3	24.2	22.2	22.9	23.1	22.1	22.9	16.8
XSUM	4.2	3.8	3.9	3.8	4.3	4.0	3.2	2.4
WebNLG EN	9.9	10.1	10.0	10.5	9.5	9.9	10.7	5.2
WikiLingua EN	4.3	4.0	4.1	3.7	4.3	4.2	4.0	2.7
bAbI	20.4	20.6	21.7	21.4	21.1	21.3	19.4	10.7
Average	23.1	23.4	23.6	23.7	24.4	23.8	21.4	15.9

M Detailed Code Augmentation Results

We report tabular results for replacing part of C4 or OSCAR with code for 4.2 billion parameter and 2.8 billion parameter models in Tables 10-11. We find that training on up to 50% of Python data maintains performance on all natural language tasks while enabling huge performance gains on state-tracking (bAbI) for C4. For OSCAR gains are less clear, which is likely due to OSCAR containing code [83], while code data was explicitly filtered out for C4 [90].

Table 10: **Results for code-augmentation for 4.2B parameter models.** Models trained on a mix of natural language (C4) and Python (The Stack). Scores are normalized averages of 0-5 few-shots and reported as percentages. We report mean/std. err. across five different models, each trained with a different random seed.

Dataset (↓)	% of Python pre-training data (remainder is C4)									
	0	10	20	30	40	50	60	70	80	90
ANLI R1	-1.0 ± 0.3	-0.7 ± 0.6	0.5 ± 0.6	-0.2 ± 1.1	-1.5 ± 0.7	-0.7 ± 1.1	-1.1 ± 0.9	-1.2 ± 1.1	-1.4 ± 0.7	-1.4 ± 0.6
ANLI R2	0.8 ± 0.5	0.4 ± 0.8	0.3 ± 1.0	0.6 ± 0.5	0.5 ± 0.6	0.3 ± 0.6	0.8 ± 0.7	0.4 ± 0.5	0.1 ± 1.2	1.0 ± 0.3
ANLI R3	1.1 ± 0.7	0.6 ± 0.5	0.5 ± 0.6	0.2 ± 0.4	0.3 ± 0.5	0.2 ± 0.5	0.3 ± 0.2	-0.1 ± 0.3	-0.0 ± 0.2	-0.1 ± 0.2
ARC-Challenge	5.3 ± 0.6	6.4 ± 1.0	5.2 ± 2.4	4.3 ± 1.4	5.2 ± 0.8	5.2 ± 0.5	2.6 ± 0.4	1.7 ± 0.5	-0.4 ± 0.4	-3.0 ± 0.4
ARC-Easy	49.2 ± 0.9	52.4 ± 1.1	49.6 ± 3.7	48.1 ± 4.1	50.1 ± 1.0	49.7 ± 0.3	48.0 ± 0.5	45.6 ± 0.5	43.3 ± 0.4	37.7 ± 0.7
BoolQ	18.2 ± 4.0	10.5 ± 12.0	16.3 ± 5.2	17.8 ± 3.3	13.4 ± 3.4	14.8 ± 2.1	12.5 ± 8.9	12.1 ± 6.6	7.2 ± 6.7	10.7 ± 7.3
CB	12.0 ± 7.2	20.3 ± 2.7	14.4 ± 7.1	16.5 ± 1.2	22.3 ± 3.1	22.1 ± 4.8	19.4 ± 4.8	23.8 ± 3.3	23.8 ± 4.1	23.4 ± 2.4
COPA	59.1 ± 5.4	56.4 ± 4.9	46.7 ± 8.7	50.2 ± 3.7	52.7 ± 2.5	50.1 ± 4.5	46.5 ± 1.9	43.1 ± 4.2	39.2 ± 3.7	35.9 ± 4.9
HellaSwag	27.8 ± 4.8	29.4 ± 0.4	25.7 ± 4.8	27.0 ± 1.7	26.3 ± 2.4	26.3 ± 0.6	25.0 ± 0.1	22.6 ± 0.1	19.5 ± 0.2	14.7 ± 0.1
PiQA	50.6 ± 0.5	50.8 ± 0.6	48.6 ± 3.0	48.2 ± 2.8	48.7 ± 0.7	48.4 ± 1.0	47.1 ± 0.7	45.6 ± 0.3	43.4 ± 0.9	39.0 ± 0.8
RTE	5.6 ± 3.1	7.3 ± 3.4	4.4 ± 4.7	6.1 ± 2.6	9.1 ± 4.0	8.1 ± 5.9	7.7 ± 5.3	4.0 ± 2.1	6.2 ± 2.1	4.6 ± 2.5
SciQ	84.6 ± 3.9	87.1 ± 0.2	84.6 ± 4.8	86.9 ± 1.2	86.9 ± 1.2	87.9 ± 0.9	87.6 ± 0.6	87.0 ± 0.2	86.0 ± 0.2	84.5 ± 0.6
StoryCloze 2016	61.1 ± 3.7	62.0 ± 0.6	59.0 ± 4.8	60.8 ± 1.5	59.9 ± 1.9	60.0 ± 0.7	59.0 ± 0.4	57.2 ± 0.5	54.9 ± 0.4	51.0 ± 0.3
WinoGrande XL	17.0 ± 2.6	17.4 ± 2.1	14.9 ± 4.4	15.2 ± 2.0	15.7 ± 1.2	14.2 ± 1.0	13.5 ± 1.3	10.7 ± 1.3	9.1 ± 0.6	5.3 ± 1.3
E2E NLG	18.2 ± 1.2	21.8 ± 1.6	15.9 ± 8.6	23.3 ± 0.6	21.5 ± 3.8	23.9 ± 0.6	23.7 ± 0.6	23.7 ± 0.5	24.3 ± 0.7	24.0 ± 0.9
XSUM	2.9 ± 0.2	3.2 ± 0.5	3.4 ± 0.3	3.3 ± 0.3	3.6 ± 0.6	3.4 ± 0.2	3.5 ± 0.2	2.9 ± 0.3	2.8 ± 0.4	2.7 ± 0.2
WebNLG EN	4.8 ± 2.0	9.5 ± 0.7	10.2 ± 1.1	10.5 ± 0.7	10.4 ± 0.8	10.4 ± 0.6	9.9 ± 0.4	10.0 ± 0.5	9.3 ± 0.6	9.2 ± 0.2
WikiLingua EN	3.3 ± 0.5	4.0 ± 0.1	4.0 ± 0.2	4.2 ± 0.1	4.3 ± 0.3	4.2 ± 0.2	4.4 ± 0.3	4.1 ± 0.2	3.9 ± 0.2	3.6 ± 0.3
bAbI	0.0 ± 0.0	12.5 ± 6.7	13.8 ± 7.2	15.8 ± 8.2	17.4 ± 9.2	23.2 ± 1.2	23.4 ± 2.0	24.3 ± 1.4	23.2 ± 1.0	24.6 ± 1.8
Average	22.1 ± 1.7	23.7 ± 0.7	22.0 ± 3.0	23.1 ± 1.1	23.5 ± 1.0	23.8 ± 0.5	22.8 ± 1.0	22.0 ± 0.6	20.8 ± 0.5	19.3 ± 0.3
Average (no bAbI)	23.4 ± 1.8	24.4 ± 0.7	22.5 ± 2.8	23.5 ± 0.8	23.9 ± 0.9	23.8 ± 0.5	22.8 ± 1.0	21.8 ± 0.6	20.6 ± 0.6	19.1 ± 0.4